

Experiments

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Abstract

This note presents an experimental evaluation of the EO-Pattern layer, an edge-conditioned instance of the MPNN architecture (Gilmer et al. [4]) in which each pairwise interaction on the hypergraph is conditioned on two quantities computed directly from hyperedge incidence: the equal-edges transition kernel P^{EE} and a cardinality descriptor χ_{vu} [2], rather than from the clique-expanded graph. Three experiments test whether χ_{vu} accesses structure that a clique expansion destroys, whether this transfers to real co-citation data, and whether P^{EE} and χ_{vu} can be jointly load-bearing within the layer’s normalized factorization. A final analysis identifies the conditions under which each quantity is informative or inert.

In a controlled synthetic graph-classification task, twin hypergraphs share a clique expansion and node features, so any clique-only model is provably at chance. Clique-based GCN/GIN and EO-Pattern ablations that ignore χ_{vu} reach 50.0% accuracy, while variants using χ_{vu} reach 100.0% accuracy and macro-F1, collapsing back to chance once χ_{vu} is shuffled – a mechanism-validation result, not a claim of general superiority.

On the Cora and Citeseer co-citation hypergraphs, under an architecture-matched comparison (same residual connection, depth, and width), the result is honestly negative: shuffling χ_{vu} does not hurt the full model, all ablations lie within one standard deviation of each other, and a residual clique-expansion GCN is clearly stronger on Cora and statistically comparable on Citeseer. Neither P^{EE} nor χ_{vu} adds signal beyond plain neighborhood aggregation on this data.

To test whether P^{EE} and χ_{vu} can be jointly load-bearing, we construct a node-classification gadget in which a target vertex has neighbor types that are indistinguishable by χ_{vu} or by P^{EE} alone, but jointly unique. The full EO-Pattern layer reaches 79.5% accuracy, a super-additive gain over the uniform EO baseline (+17.0 pp, vs. +5.6 pp for P^{EE} alone and +1.7 pp for χ_{vu} alone), and randomizing χ_{vu} collapses it to the P^{EE} -only level. Joint usefulness is constructible, not typical.

Two structural propositions explain when this can happen: χ_{vu} is constant on k -uniform hypergraphs, and P^{EE} reduces to uniform neighborhood averaging on k -uniform linear hypergraphs; together with a third, task-dependent regime in which the quantities vary but are not predictively aligned with the labels, as suggested by the co-citation results. Together these conditions reconcile the positive synthetic result with the negative real-data result and motivate a concrete, falsifiable criterion for predicting, a priori, when the EO-Pattern quantities are likely to help.

Code availability. The code used to reproduce all experiments is available at

<https://github.com/adriruiz12/bachelor-thesis-eompp.git>.

The repository is organized as a reproducibility package rather than as a polished software library: the shared implementation of the EO-Pattern layer is contained in `eompp/`, while the three experimental pipelines are separated into `Synthetic/`, `Benchmark/`, and `Complementarity/`. At the time of writing, the repository uses descriptive experimental folder names and neutral module names to make the role of each component explicit. The final public version will include the definitive metadata, license information, and author information.

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1 Synthetic Hypergraph Classification Task

We evaluate our EO-Pattern MPNN on a synthetic hypergraph classification task designed to isolate the effect of hyperedge cardinality. Each example consists of a pair of hypergraphs built on the same base graph: one class (“big”) uses the full cliques as hyperedges, while the other class (“small”) replaces each clique by a finer decomposition (here, all of its 2-element subsets). By construction, the two hypergraphs have identical clique expansions, so any model relying only on the graph projection (e.g. standard GCN or GIN) receives identical inputs for both classes. Thus graph-based baselines provably perform at chance (50%). In contrast, our EO-Pattern layer incorporates the cardinality descriptor χ (the sizes of shared hyperedges) in its message function, enabling it to distinguish the classes. We generate 300 random base graphs (with clique sizes and overlaps as in our configuration) and split them by base graph into training, validation and test sets (no graph appears in more than one split).

We compare several models: two standard graph neural networks applied to the clique-expansion of each hypergraph (Clique-GCN, Kipf and Welling [7]; and Clique-GIN, Xu et al. [8]), and ablations of our EO-Pattern architecture. In particular: (A) a uniform pairwise MPNN baseline with the same vertex update architecture as EO-Pattern, but with both hypergraph-derived ingredients removed, so that it aggregates uniformly over adjacent vertices and does not use P^{EE} or χ ; (B) EO-Pattern with equal-edge random-walk weights, but ignoring χ ; (C) using only the χ descriptor with uniform propagation weights; (D) the full EO-Pattern MPNN, using both weights and χ ; and (E) a shuffled- χ control variant. All models are implemented as 2-layer MPNNs (Gilmer et al. [4]) with 64 hidden units and trained for up to 300 epochs (with early stopping), and results are averaged over 5 random seeds. Table 1 reports the test accuracy and macro F1¹ score for each model (mean $\pm$ std).

The results validate the intended mechanism in this controlled construction. All models that ignore the hyperedge cardinalities (the clique-based GCN/GIN and variants (A), (B), (E)) score 50% accuracy (random guessing) as expected. By contrast, the EO-Pattern variants that leverage χ ((C) and (D)) achieve 100% accuracy and F1, perfectly separating the classes. This confirms that,

¹See Appendix A.1 for the role of each metric in this experiment.

Table 1: performance on the synthetic hypergraph classification task (mean $\pm$ std over 5 runs).

Models (C) and (D) use the true cardinality descriptor χ and correctly distinguish the two classes, achieving perfect accuracy. All other models, which see only the clique-expanded graph (or random χ), perform at chance.

Model	Accuracy (%)	F1-score (%)
Clique-GCN (graph MPNN)	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
Clique-GIN (graph MPNN)	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(A) Uniform EO baseline	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(B) P^{EE} only	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0
(C) χ only	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
(D) EO-Pattern (full model)	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
(E) Randomized χ	50.0 $\pm$ 0.0	33.3 $\pm$ 0.0

in this controlled twin construction, the discriminating signal is exactly the cardinality information encoded by χ_{vu} : models that do not receive this descriptor cannot separate the two classes, while models that receive it can. In summary, the synthetic experiment demonstrates that EO-Pattern MPNN can exploit cardinality-sensitive hypergraph-derived connectivity information that is invisible to unweighted clique-expansion baselines.

2 Real-World Benchmark: Co-citation Node Classification

The synthetic experiment of Section 1 establishes that the EO-Pattern layer can access structure that clique-expansion models provably lose. We now test whether this advantage transfers to real-world data: node classification on the Cora and Citeseer co-citation hypergraphs (inspired by Yadati et al. [9]), in which each hyperedge groups a paper with the set of papers it cites. We use the random per-class 50%/25%/25% train/val/test splits of Chien et al. [1], one per seed, and report test accuracy as mean $\pm$ std.

The co-citation hypergraphs were constructed using **TopoNetX**, the topological-domain construction package from the TopoX suite of Hajij et al. [5]. In the implementation, each citation hyperedge is added as a rank-one cell of a **ColoredHyperGraph**, and the node-hyperedge incidence matrix is obtained through `incidence_matrix(0,1)`. Since **TopoNetX** fixes an internal node order, the rows of the incidence matrix are reordered to match the canonical node indexing used by the feature and label matrices. The EO quantities P^{EE} and χ_{vu} are then computed from this incidence matrix. A dependency-free **SciPy** incidence builder is kept as a fallback and cross-check, and the two incidence matrices are asserted to agree.

Nine models are evaluated: a feature-only MLP control, **Clique-GCN**, **Clique-GIN**, **AllDeepSets**, and the five EO-Pattern ablations (A)-(E) described in Section 1. **AllDeepSets** is our pure-PyTorch reimplementation² of the Deep Sets propagation variant of Chien et al. [1], included as the hypergraph-native incidence baseline. All structural node-classification models share the same per-layer residual connection (He et al. [6] $h \leftarrow \text{relu}(\text{out}) + h$), two layers, and hidden width

²**AllDeepSets** is reimplemented in pure PyTorch so that it is evaluated inside the same experimental pipeline as the other models, with the same input projection, residual structure, dropout, training loop, splits, early stopping, and metrics. It is included as the hypergraph-native incidence baseline, not as a benchmark of the official AllSet implementation.

128. The feature-only MLP is kept as a non-residual control.

Table 2: node-classification accuracy (%) on Cora and Citeseer co-citation hypergraphs (mean $\pm$ std over 20 seeds; hidden 128, 2 layers, 50%/25%/25% splits). *all*: accuracy over all test nodes; *active*: accuracy restricted to nodes with $d_H(v) > 0$, i.e. nodes belonging to at least one hyperedge of size at least two (53% of Cora and 44% of Citeseer nodes are active).

Model	Cora		Citeseer	
	all	active	all	active
MLP (features only)	73.8 $\pm$ 1.4	75.1 $\pm$ 2.0	72.2 $\pm$ 0.9	74.9 $\pm$ 1.9
Clique-GCN	79.8 $\pm$ 1.7	84.8 $\pm$ 1.7	73.3 $\pm$ 1.3	76.1 $\pm$ 2.0
Clique-GIN	76.5 $\pm$ 1.4	82.8 $\pm$ 1.5	71.3 $\pm$ 1.0	73.9 $\pm$ 2.0
AllDeepSets	75.2 $\pm$ 1.5	76.5 $\pm$ 2.0	73.1 $\pm$ 1.0	75.9 $\pm$ 1.9
(A) Uniform EO baseline	76.2 $\pm$ 1.2	80.3 $\pm$ 1.4	72.4 $\pm$ 1.1	75.7 $\pm$ 1.6
(B) P^{EE} only	76.6 $\pm$ 1.3	80.9 $\pm$ 1.6	72.5 $\pm$ 1.1	76.0 $\pm$ 1.5
(C) χ only	76.2 $\pm$ 1.5	80.4 $\pm$ 2.2	72.6 $\pm$ 1.1	76.0 $\pm$ 1.8
(D) EO-Pattern (full model)	76.3 $\pm$ 1.4	80.8 $\pm$ 1.8	72.5 $\pm$ 0.9	76.2 $\pm$ 1.6
(E) Randomized χ	76.6 $\pm$ 1.3	80.8 $\pm$ 1.9	72.6 $\pm$ 1.1	76.1 $\pm$ 1.6

Under this matched comparison, **Clique-GCN** is clearly stronger than the EO variants on Cora and statistically comparable on Citeseer. The co-citation data contain a small number of repeated hyperedges (96 of 1579 columns on Cora, 75 of 1079 on Citeseer), corresponding to identical vertex sets induced by distinct citations. These repetitions are kept as multiplicities, consistently with treating $\mathcal{E}$ as a family of hyperedges with multiplicities. They leave the adjacency support $N_H(v)$ unchanged, but they can affect the values of $d_H(v)$, $P^{\text{EE}}(v, u)$, and χ_{vu} . Since this affects all EO-Pattern variants equally, the negative conclusion below is unaffected.

The honest reading of Table 2 is a negative result for χ on co-citation data. On both datasets the shuffled- χ control (E) matches the full model (D), so the specific cardinality assignment carries no usable signal here. On Cora, the ablation ladder (A)-(D) spans 76.2 $\rightarrow$ 76.6 with per-model standard deviation between 1.2 and 1.5. All differences lie within one standard deviation, and $(E) = 76.6 \approx (D) = 76.3$ confirms χ is inert. The approximate 2.4-point gain of EO models over the MLP (73.8) is consistent across all 20 seeds (gap exceeds one standard deviation) and attributable to neighborhood aggregation already present in (A); neither P^{EE} nor χ add anything on top. Note, however, that variant (A) reaches 76.2, which is clearly below the residual **Clique-GCN** result of 79.8. Thus, on Cora, the EO-Pattern aggregation operator is not competitive with the tuned residual GCN baseline. This should be separated from the ablation question addressed by the (A)-(E) ladder: whether P^{EE} and χ add value within the EO-Pattern family itself. The nearly flat ladder suggests that, on this dataset, they do not provide a clear additional gain. On Citeseer, all EO-Pattern variants are also indistinguishable from the MLP.

Another observation concerns node coverage: 47% of Cora nodes and 56% of Citeseer nodes lie in no hyperedge of size ≥ 2 (formally, $d_H(v) = 0$ in the notation of Section 4) and receive no hypergraph propagation under any structural model, which affects all models' *all*-node numbers similarly and does not differentially favor EO over the explicit clique-expansion baselines. This reflects the co-citation hyperedge construction of Yadati et al. [9]: each hyperedge groups a paper with the papers it cites among the dataset, and hyperedges with fewer than two members are dropped. A node is isolated precisely when it cites no other in-dataset paper and is not itself cited within any such

group.

This does not contradict the result of Section 1; it sharpens the question. The failure on co-citation does not settle whether the EO quantities are ever useful for node classification. The following section constructs a case in which the normalized quantities P^{EE} and χ are jointly useful, and in which their combination identifies information that either quantity alone loses.

3 Joint Usefulness of P^{EE} and χ : A Constructive Gadget

The benchmark of Section 2 shows that neither P^{EE} nor χ helps on co-citation data. A natural question is whether the two quantities are ever jointly load-bearing at all. We answer affirmatively with a constructive synthetic experiment: we design a node-classification task in which the informative neighbor is identifiable from the joint signature (P^{EE}, χ) , but not from either normalized quantity alone.

Each gadget attaches to a target vertex v three types of neighbor, distinguished by the way they connect to v :

- A single informative neighbor (Q), carrying the true label of v , connected through two size-3 hyperedges.
- m decoys of type P , carrying random labels, each connected through one size-2 hyperedge.
- m decoys of type R , carrying random labels, each connected through one size-3 hyperedge.

with neutral filler vertices completing the size-3 hyperedges. These filler vertices are also adjacent to v , but they carry neutral features and serve only to complete the hyperedges. Writing $Z_{vu} = \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e|-1}$ for the equal-edges mass shared by v and u , the three neighbor types have the signatures shown in Table 3.

Table 3: signatures of the three neighbor types. The descriptor χ separates P (size 2) from $\{Q, R\}$ (size 3) but cannot distinguish Q from R . The kernel P^{EE} separates R ($Z_{vu} = 1/2$) from $\{P, Q\}$ ($Z_{vu} = 1$) but cannot distinguish P from Q . Only the joint signature (size 3, $Z_{vu} = 1$) is unique to Q .

Type	shared cardinality	Z_{vu}	distinguished by
P	2	1	χ
Q	3	1	– (target)
R	3	1/2	P^{EE}

By construction, neither normalized quantity alone identifies Q : χ collapses Q and R , while P^{EE} collapses P and Q . Only their combination yields a unique signature for the informative neighbor among the P/Q/R types. This is the precise sense in which the experiment tests complementarity. We stress that the claim is relative to the normalized factorization (P^{EE}, χ) adopted by the layer: the unnormalized binned equal-edges mass, namely the vector of Z_{vu} values per cardinality bin before normalization by Z_{vu} , would already yield distinct signatures $P = (1, 0)$, $Q = (0, 1)$, and $R = (0, 1/2)$. Therefore, the gadget does not prove that no single hypergraph-derived descriptor can solve the task; it rather shows that the two normalized quantities used by this layer encode complementary information.

The full model (D) reaches 79.5%, well above both the P^{EE} -only ablation (B, 68.1%) and the χ -only ablation (C, 64.2%). The important point is the interaction effect: P^{EE} alone adds +5.6 pp

Table 4: performance on the P/Q/R complementarity gadget (mean $\pm$ std over 10 seeds; 450 gadgets, $m = 4$ decoys per type, hidden 64, 1 layer). The full model (D) is well above either single-ingredient ablation (B), (C), and randomizing χ (E) collapses performance back to the level of P^{EE} alone (B). AllDeepSets returns an identical 54.9% across all 10 seeds (std 0), i.e. it collapses to a constant majority-class predictor. Under the shared hyperparameter budget it was not tuned separately from the other models, so this entry is reported as a non-tuned reference rather than as evidence of an expressivity limit of the AllDeepSets architecture

Model	Accuracy (%)
MLP (features only)	54.2 ± 2.1
Clique-GCN	63.5 ± 3.8
Clique-GIN	63.3 ± 3.5
AllDeepSets	54.9 ± 0.0
(A) Uniform EO baseline	62.5 ± 3.9
(B) P^{EE} only	68.1 ± 3.2
(C) χ only	64.2 ± 4.6
(D) EO-Pattern (full model)	79.5 ± 3.0
(E) Randomized χ	68.6 ± 3.5

over the uniform EO baseline (A, 62.5%), χ alone adds only +1.7 pp, yet their combination adds +17.0 pp. The control (E) falls back to the P^{EE} -only level ($68.6 \approx \text{EO-B } 68.1$), indicating that the improvement in (D) is driven by the genuine χ assignment acting jointly with P^{EE} , not simply by adding parameters.

Two points limit the scope of this result. First, the accuracy floor is above chance: because the informative neighbor Q is always present and truthful, even uniform aggregation extracts a weak signal. The relevant comparison is the gap between (D) and the ablations, not the absolute level. Second, this is a constructive existence argument, not an empirical generalization claim: it shows that a regime of joint usefulness exists within this normalized decomposition, not that such regimes are common in natural data. Third, Q 's feature is an input available at inference for every split and does not constitute leakage: test labels enter neither the loss nor the early-stopping criterion. The task is to identify Q among decoys that match it on either single quantity alone, which is why both single-ingredient ablations (B, C) and the randomized- χ control (E) fail to reach the level of (D), and even (D) reaches only 79.5%, not 100%.

4 When Do the Hypergraph-Derived Quantities Carry Signal?

The three experiments form a consistent picture only if we can say precisely when P^{EE} and χ are informative and when they are inert. We do not have a full (necessary and sufficient) characterization, because sufficiency is task-dependent. We do, however, have two clean necessary structural conditions, which we prove directly from the definitions. Recall that, for an adjacent pair v, u ,

$$Z_{vu} = \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1}, \quad P^{\text{EE}}(v, u) = \frac{Z_{vu}}{d_H(v)}, \quad \chi_{vu} = \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v,u)} \frac{1}{|e| - 1} \mathbf{c}_{|e|},$$

where $\mathbf{c}_r$ is the clamped one-hot cardinality encoding defined by the same `max_card` convention as in the layer implementation.

Proposition 4.1 (Structural inertness of χ). *If H is k -uniform on its hyperedges of size at least two (i.e.: all such hyperedges have cardinality k), then $\chi_{vu} = \mathbf{c}_k$ for every adjacent pair (v, u) . Consequently χ is a constant feature and carries no information. Conversely, χ is non-constant only if H contains hyperedges belonging to at least two distinct cardinality bins.*

Proof. Each $e \in \mathcal{E}(v, u)$ contributes the term $\frac{1}{|e|-1} \mathbf{c}_{|e|}$. If every such e has $|e| = k$, then all contributions point along $\mathbf{c}_k$, so

$$\chi_{vu} = \frac{1}{Z_{vu}} \left(\sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e|-1} \right) \mathbf{c}_k = \frac{1}{Z_{vu}} Z_{vu} \mathbf{c}_k = \mathbf{c}_k,$$

independent of the pair. The converse follows by contrapositive. If all hyperedges of size at least two have the same cardinality k , then the preceding computation shows that every adjacent pair has the same descriptor $\mathbf{c}_k$. Therefore, if χ is non-constant, the hypergraph cannot be k -uniform, so it must contain hyperedges of at least two distinct cardinality bins. $\square$

Proposition 4.2 (Collapse of P^{EE} to the uniform mean). *For any vertex v , the row $P^{\text{EE}}(v, \cdot)$ is uniform over the neighborhood $N_H(v)$ if and only if Z_{vu} is constant over $u \in N_H(v)$. In particular, if H is k -uniform and linear (any two vertices lie in at most one common hyperedge), then $P^{\text{EE}}(v, u) = 1/|N_H(v)|$ for all u , i.e. P^{EE} coincides exactly with uniform neighborhood averaging and provides no signal beyond it.*

Proof. If $Z_{vu} = z$ for all $u \in N_H(v)$, then $P^{\text{EE}}(v, u) = z/d_H(v)$ is constant across the row. Since the row is a probability distribution over $|N_H(v)|$ neighbors, the constant equals $1/|N_H(v)|$. Conversely, a uniform row forces $Z_{vu} = d_H(v)/|N_H(v)|$, constant. For the linear k -uniform case, each pair shares a single hyperedge of size k , so $Z_{vu} = \frac{1}{k-1}$; linearity makes the neighbor sets of distinct hyperedges through v disjoint, so $|N_H(v)| = (k-1)d_H(v)$, and $P^{\text{EE}}(v, u) = \frac{1}{k-1}/d_H(v) = 1/|N_H(v)|$. $\square$

Propositions 4.1 and 4.2 identify two structural regimes in which the hypergraph-derived quantities are inert by construction. Together with a third, task-dependent regime, they explain all three experiments:

1. **Structural degeneracy.** When H is k -uniform (Proposition 4.1) or k -uniform and linear (Proposition 4.2), χ or P^{EE} reduce to constants and cannot help. This is a property of the hypergraph alone, independent of the task.
2. **Feature degeneracy.** When node features are constant, any row-stochastic aggregation weight maps constant inputs to constant outputs, so P^{EE} is inert regardless of its values. This is why ablation (B) sits at chance in Section 1, even though P^{EE} there does differ between the two classes: only χ , which enters the message function as a feature, can break the node symmetry.
3. **Lack of predictive alignment.** When H is non-degenerate (so χ and P^{EE} vary) but this variation is not aligned with the target labels, both quantities are non-constant yet unhelpful. This is the co-citation regime of Section 2: the datasets avoid the degeneracies of Propositions 4.1 and 4.2 (their hyperedges have several distinct sizes and are not linear, so χ and P^{EE} are genuinely non-constant), but their variation does not provide useful predictive signal under the present protocol.
4. **Vacuous case (isolated vertices).** For v with $d_H(v) = 0$ (47%/56% of Cora/Citeseer nodes, Section 2), $N_H(v) = \emptyset$, so neither χ_{vu} nor $P^{\text{EE}}(v, u)$ is defined for any u : the EO-Pattern layer receives a zero aggregated message and reduces to an MLP-style update. This

is orthogonal to the three regimes above (it is a property of the vertex’s incidence, not of χ or P^{EE}) but it is the dominant source of the all-node/active-node gap in Table 2 for every model, EO or not.

The complementarity gadget of Section 3 is, by contrast, a regime in which all three degeneracies are avoided and the structural variation is engineered to correlate with the label, so both quantities contribute within the chosen normalized factorization. This shows that the regime of joint usefulness is non-empty, but a general characterization of when structural variation aligns with labels remains an open problem.

5 Discussion and Future Work

The experiments and Propositions 4.1 and 4.2 together support a precise and honest reading of the proposal. The EO-Pattern layer is well defined and permutation-equivariant, its cardinality descriptor provably accesses structure that clique expansion destroys (Section 1), and there exist tasks in which, within the layer’s normalized factorization, the equal-edges kernel and the descriptor provide complementary information through their joint signature (Section 3). On standard co-citation benchmarks, however, neither quantity yields a gain beyond plain neighborhood aggregation (Section 2). The propositions explain why a positive synthetic result and a null real-world result are not in tension: the necessary structural conditions are met on co-citation, but the missing ingredient appears to be predictive alignment between cardinality structure and labels, which is data-dependent and cannot be established from structure alone.

This identifies the central open problem: a sufficient condition for the EO-Pattern quantities to help. We propose a concrete, measurable direction. For a labeled hypergraph one can compute, over the set of adjacent pairs, the mutual information between the descriptor χ_{vu} (or the scalar Z_{vu}) and the label-agreement indicator $\mathbb{1}[y_v = y_u]$. When this mutual information is near zero, the task lies in the low-alignment regime, so inertness can be predicted a priori, without training. Such a statistic would turn the present negative co-citation result from a post-hoc observation into a falsifiable prediction, and would provide a principled criterion for selecting datasets on which the EO-Pattern layer is expected to be beneficial.

Three further directions follow naturally. First, evaluation in the label-scarce regime: structural propagation tends to matter more when labeled data is scarce, so the sparse semi-supervised splits of HyperGCN [9] (around twenty labels per class) are a natural complement to the data-rich 50%/25%/25% protocol used here. Second, the curvature-gated extension outlined in the proposal, replacing $P^{\text{EE}}(v, u)$ by $P^{\text{EE}}(v, u) g(\kappa_{EE}(v, u))$ with an Ollivier–Ricci curvature gate, which would make the layer sensitive to local hypergraph geometry beyond cardinality and may be effective in regimes where the present quantities cannot. Third, the benchmark could be strengthened by including additional incidence-based hypergraph neural networks such as HGNN [3] or AllSet-Transformer [1]; we include AllDeepSets as the representative incidence baseline and defer broader comparison to future work.

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A Experimental Protocol and Reproducibility Details

This appendix summarizes the experimental conventions used throughout the three experiments.

A random seed fixes the pseudo-random state for weight initialization, data splits, and stochastic optimization. The role of seeds differs across experiments:

- **Synthetic.** The dataset and the train/val/test split are fixed (`data_seed=0`, `split_seed=0`). The five seeds vary only model weight initialization. Because the result is provably determined by the construction (50 % or 100 % by design), this is sufficient.
- **Benchmark.** Each of the twenty seeds draws an independent per-class 50 %/25 %/25 % split, so the mean and standard deviation reflect both split variation and initialization variance.
- **Complementarity gadget.** The hypergraph structure is fixed (`seed=12345` in the generator). Each of the ten seeds draws an independent split over target nodes and re-initializes the model weights.

Two metrics are used. Accuracy is the fraction of correctly classified examples. Macro-F1 is the unweighted average of the F1 score across classes, and is included to detect degenerate classifiers that obtain reasonable accuracy by overpredicting one class. This explains why chance-level accuracy can be accompanied by a macro-F1 score below 50% in balanced binary experiments: a model that collapses to predicting a single class can classify roughly half of the examples correctly while having poor class-wise F1.

The ablation study isolates the role of the two hypergraph-derived ingredients of the EO-Pattern layer. Model (A) uses only uniform clique-style neighborhood aggregation. Model (B) uses P^{EE} but ignores χ_{vu} . Model (C) uses χ_{vu} but replaces P^{EE} by uniform weights. Model (D) is the full EO-Pattern layer. Model (E) is a negative control in which χ_{vu} is randomly shuffled across adjacent pairs. Therefore, if (E) performs similarly to (D), the specific assignment of cardinality descriptors is not carrying useful signal for the task.

All experiments are designed to answer a different question. The synthetic twin experiment tests whether χ_{vu} can recover information provably erased by clique expansion. The co-citation benchmark tests whether this advantage transfers to standard real-world node-classification data. The P/Q/R gadget tests whether P^{EE} and χ_{vu} can provide complementary information in a controlled setting. The three experiments should therefore be read together: the first validates the mechanism, the second gives an honest negative benchmark, and the third shows that a regime of joint usefulness within the normalized factorization can be constructed.

A.1 Accuracy and macro-F1.

The main metric reported in the experiments is accuracy. For a test set $\mathcal{T}$, accuracy is defined as

$$\text{Acc} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbf{1}\{\hat{y}_i = y_i\}.$$

It measures the fraction of correctly classified examples and is the standard metric used in node-classification benchmarks such as Cora and Citeseer.

In addition, we report macro-F1 in the synthetic twin experiment. For each class c , the F1 score is the harmonic mean of precision and recall,

$$F1_c = 2 \frac{\text{Prec}_c \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c},$$

and macro-F1 is the unweighted average over classes,

$$\text{MacroF1} = \frac{1}{C} \sum_{c=1}^C F1_c.$$

Unlike accuracy, macro-F1 penalizes classifiers that ignore one or more classes. This is relevant in the synthetic experiment because several baselines achieve chance-level accuracy by collapsing to essentially one class. In a balanced binary task, such a classifier may still obtain around 50% accuracy, but its macro-F1 drops to about 33.3%, revealing that it has not learned a meaningful class distinction.

For this reason, macro-F1 is most informative in the synthetic experiment, where the purpose is to verify whether the model truly distinguishes the two hypergraph classes rather than exploiting a degenerate prediction strategy. In the co-citation benchmark, accuracy is kept as the main reported metric because it is the conventional metric for Cora and Citeseer comparisons; macro-F1 is computed as an auxiliary diagnostic but does not change the qualitative conclusions. In the complementarity experiment, the task is a controlled and approximately balanced binary classification problem, so accuracy already captures the intended success criterion: whether the model can solve the P/Q/R gadget or not.